VITyarthi | Open Source Software (CSE0002)

The Open Source Audit

A Capstone Project for OSS NGMC Course

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Introduction :

Open Source Software (OSS) has emerged as a powerful paradigm in modern software development, transforming the way software is created, distributed, and maintained. Unlike proprietary software, where access to the source code is restricted, open-source software provides users with the freedom to access, modify, and distribute the code. This openness promotes collaboration, transparency, and rapid innovation.

In today's technology-driven world, open-source software plays a crucial role in fields such as web development, artificial intelligence, cloud computing, and system administration. Many widely used platforms, including Linux and Python, are built on open-source principles.

This project focuses on understanding open-source concepts through the practical implementation of Python in a Linux environment. The project was carried out using Ubuntu on Windows Subsystem for Linux (WSL), which enables users to run a Linux environment directly within a Windows system without requiring a virtual machine.

The primary objectives of this project are:

- To understand the origin and philosophy of open-source software
- To explore the Linux operating environment
- To analyze the Free and Open Source Software (FOSS) ecosystem
- To implement shell scripts for system monitoring and automation

This project combines theoretical understanding with practical implementation, providing a comprehensive insight into open-source technologies.

Origin And Philosophy – (Part-A) :

❖ A.1 The Problem This Software Was Created to Solve

Before the development of Python, programming languages such as C and C++ dominated the industry. While these languages were powerful and efficient, they posed several challenges. They required complex syntax, manual memory management, and extensive coding effort, making them difficult for beginners to learn and time-consuming for developers to use.

Python was developed by Guido van Rossum in 1991 to address these issues. The primary goal was to create a language that is easy to read, easy to write, and highly efficient. Python introduced a clean and simple syntax that resembles natural language, significantly reducing the complexity of programming.

Additionally, Python improved developer productivity by allowing tasks to be completed with fewer lines of code. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming, making it flexible for various applications.

Thus, Python successfully solved the problem of complexity in programming and made software development more accessible to a wider audience.

❖ A2. The License — What It Actually Says

Python is distributed under the Python Software Foundation (PSF) License, which is a permissive open-source license. This license allows users significant freedom while ensuring minimal restrictions.

Key Features of the License:

- Free usage for both personal and commercial purposes
- Permission to modify the source code
- Freedom to distribute original or modified versions

Fundamental Freedoms of Open Source:

- Freedom to run the program for any purpose
- Freedom to study how the program works
- Freedom to modify the program according to needs

- Freedom to distribute copies to others

Comparison with Other Licenses:

- GPL (General Public License): Requires that any modified software must also remain open source
- MIT License: Provides maximum flexibility and allows integration into proprietary systems

The PSF License strikes a balance between openness and flexibility, making Python suitable for both individual developers and large organizations.

❖ A3. The Ethics of Open Source

The philosophy of open source extends beyond technical aspects and is deeply rooted in ethical principles such as collaboration, transparency, and equal access to knowledge.

Open-source software enables developers to learn from existing codebases, contribute to community-driven projects, and innovate without restrictions. It removes financial barriers, allowing students and developers from all backgrounds to access high-quality tools and resources.However, with freedom comes responsibility.

Ethical use of open-source software requires:

- Respecting licensing terms
- Giving proper credit to original developers
- Avoiding misuse or unethical distribution

Open source fosters a culture of sharing and collective growth, making it a vital component of modern technological advancement.

Linux Footprint – (Part-B) **:**

The practical implementation of this project was carried out using Ubuntu on Windows Subsystem for Linux (WSL). WSL provides a fully functional Linux environment within a Windows system, eliminating the need for a separate virtual machine.

System Analysis and Observations:

- Python was installed using the apt package manager, a standard tool for managing software in Debian-based systems
- The installation was verified using the command python3 --version
- Python executables are located in system directories such as /usr/bin
- The system runs on Ubuntu 24.04 LTS with a WSL2 kernel

Linux is widely used in servers and development environments due to its stability, security, and efficiency. It provides powerful tools for scripting, automation, and system management, making it an ideal platform for open-source development.

The following screenshots confirm Python installation and configuration on the Linux system:

```
shalini_anand813@hp:~$ sudo apt install python3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
shalini_anand813@hp:~$
```

The FOSS Ecosystem – (Part-C) :

Python is an integral part of the Free and Open Source Software (FOSS) ecosystem, which consists of a vast network of developers, tools, libraries, and frameworks.

Key Characteristics of the FOSS Ecosystem:

- Large and active global developer community
- Availability of thousands of open-source libraries
- Continuous updates and improvements
- Strong documentation and community support

Major Application Areas:

- **Web Development:** Frameworks like Django and Flask
- **Data Science:** Libraries such as NumPy and Pandas
- **Artificial Intelligence:** Machine learning and deep learning applications
- **Automation:** Scripting and task automation

The FOSS ecosystem encourages innovation and collaboration, ensuring that software evolves continuously to meet modern technological demands.

Open Source Vs Proprietary – (Part-D) :

Feature	Open Source (Python)	Proprietary Software
Source Code	Available	Restricted
Cost	Free	Paid
Customization	High	Limited
Transparency	High	Low
Community Support	Strong	Limited

Open-source software provides flexibility, transparency, and cost-effectiveness. Users can modify and improve the software according to their needs.

In contrast, proprietary software offer controlled environments, better official support, and enhanced security in some cases, but limits user freedom.

For learning, experimentation, and innovation, open-source software is more advantageous.

Shell Scripts :

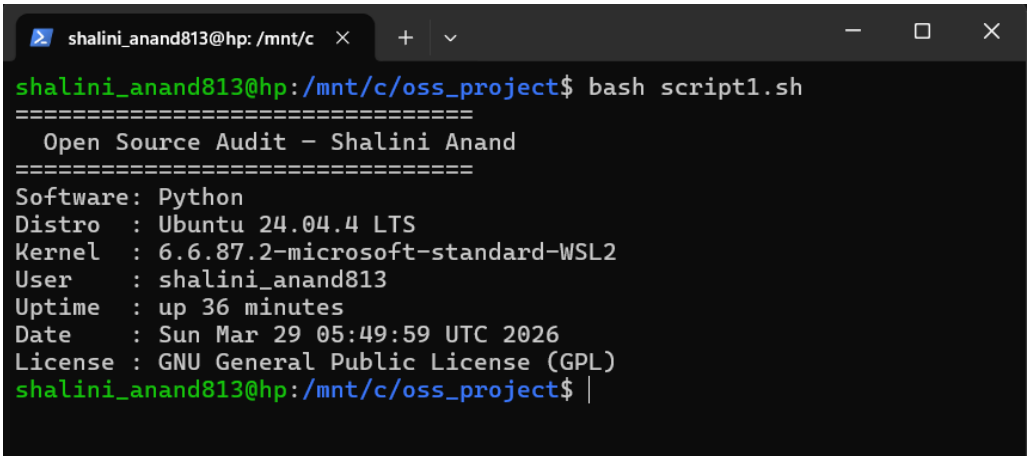
Shell scripting plays a crucial role in automating tasks and managing system operations in Linux environments. In this project, five scripts were developed to demonstrate various functionalities.

❖ *Script 1 – System Identity Report*

This script collects and displays key system information, including:

- Username
- Kernel version
- System uptime
- Linux distribution
- Current date

It provides a quick overview of the system configuration and environment.

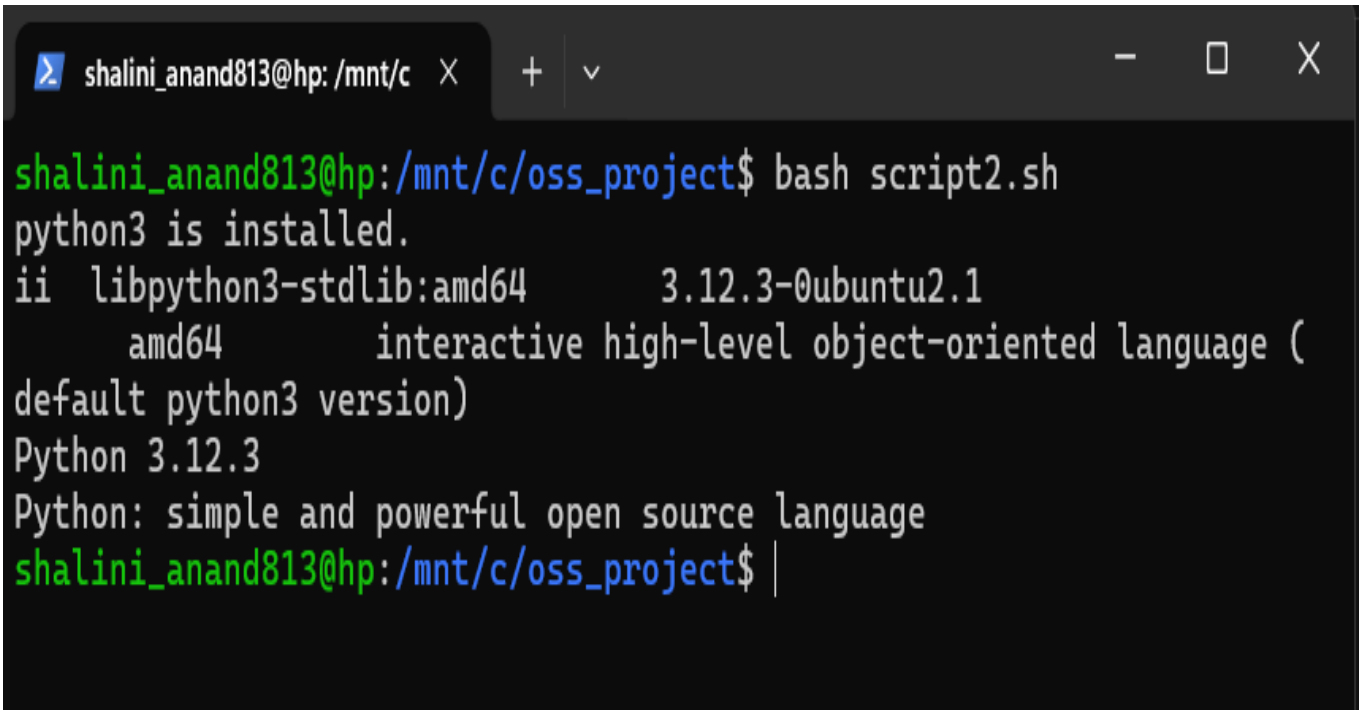


❖ *Script 2 – Package Inspector*

This script checks whether Python is installed on the system. It:

- Verifies installation status
- Displays package details
- Shows Python version

Conditional statements are used to provide appropriate output based on system status.



❖ *Script 3 – Disk and Permission Auditor*

This script analyzes important system directories and provides:

- File and directory permissions
- Ownership details
- Disk usage information.

It helps in understanding system structure and access control mechanisms.

```
shalini_anand813@hp: /mnt/c
shalini_anand813@hp:/mnt/c/oss_project$ bash script3.sh
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x root root | Size: 4.3M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 33M
/home => Permissions: drwxr-xr-x root root | Size: 44K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 95M
/tmp => Permissions: drwxrwxrwt root root | Size: 28K

Python Config Directory:
drwxr-xr-x 2 root root 4096 Feb 10 00:55 /etc/python3
shalini_anand813@hp:/mnt/c/oss_project$
```

❖ *Script 4 – Log File Analyzer*

This script scans system log files for a specific keyword such as “error”. It:

- Counts the number of occurrences
- Displays recent matching entries

```
shalini_anand813@hp: /mnt/c
shalini_anand813@hp:/mnt/c/oss_project$ bash script4.sh /var/log/dpk
g.log error

Keyword 'error' found 39 times in /var/log/dpkg.log

Last 5 matching lines:
2026-02-10 00:56:02 status unpacked liberror-perl:all 0.17029-2
2026-02-10 00:56:32 configure liberror-perl:all 0.17029-2 <none>
2026-02-10 00:56:32 status unpacked liberror-perl:all 0.17029-2
2026-02-10 00:56:32 status half-configured liberror-perl:all 0.17029
-2
2026-02-10 00:56:32 status installed liberror-perl:all 0.17029-2
shalini_anand813@hp:/mnt/c/oss_project$
shalini_anand813@hp:/mnt/c/oss_project$
```

Observation:

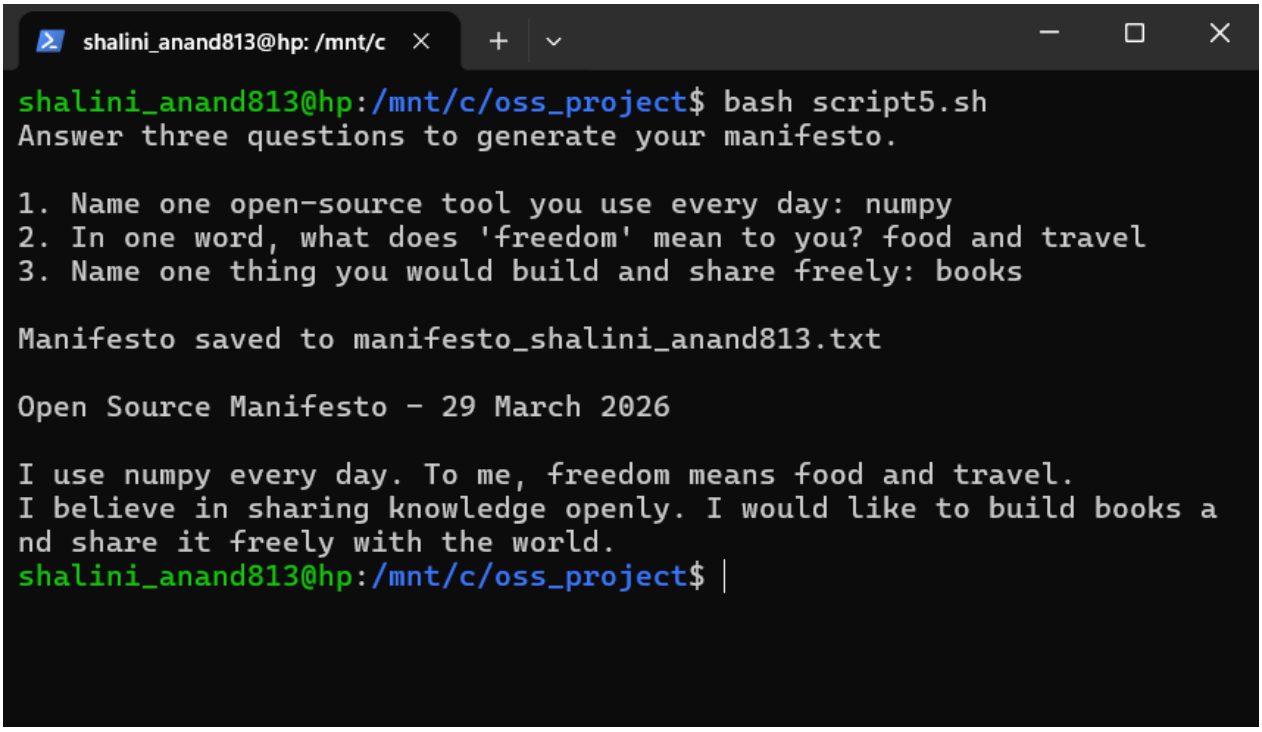
Some matches may not represent actual system errors, as the keyword can appear within package names. This highlights a limitation of simple keyword-based analysis and suggests the need for more advanced filtering techniques.

❖ *Script 5 – Manifesto Generator*

This script interacts with the user by taking input and generating a personalized open-source manifesto. It:

- Accepts user responses
- Creates a text file
- Displays the generated content
-

It demonstrates user interaction, file handling, and output formatting in shell scripting.



Project Timeline :

Day	Task
Day1	*Setup WSL and install Ubuntu*
Day2	*Install Python and explore Linux environment*
Day3	*Develop and test Script 1 and Script 2*
Day4	*Develop Script 3, Script 4, and Script 5*
Day5	*Perform testing, capture screenshots, and prepare report*

The project was completed in a structured manner over several weeks. Initially, the basic concepts of Linux and FOSS were studied to build a strong foundation. This was followed by learning Python and shell scripting techniques. In the next phase, scripts were developed and tested to ensure correct functionality. Finally, the report was prepared by organizing all theoretical and practical aspects in a systematic manner.

REFERENCES :

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- **Van Rossum, G. Python Programming Language**. : https://www.python.org
- **Free Software Foundation.** : https://www.gnu.org/philosophy/free-sw.html
- **Open Source Initiative**. : https://opensource.org/osd
- **Python Software Foundation**. : https://docs.python.org/3/license.html
- **GitHub. About Open Source and Collaboration**. : https://docs.github.com
- **The Linux Documentation Project. Linux System Documentation**. : https://tldp.org
- **GeeksforGeeks. Linux and Python Tutorials**. : https://www.geeksforgeeks.org

CONCLUSION :

This project provided comprehensive exposure to open-source software, Linux systems, and shell scripting. It demonstrated how Python can be effectively used within an open-source environment and how Linux tools can be utilized for system monitoring and automation.
The use of WSL simplified the process by enabling Linux functionality within a Windows system. Through this project, key concepts such as open-source philosophy, licensing, system operations, and scripting were understood both theoretically and practically.
Overall, the project enhanced technical skills, problem-solving ability, and understanding of modern software development practices, making it a valuable learning experience.